



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

CONSUL PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Distributed Systems

TOTAL: 10 QUESTIONS

Q1 What is Consul and what problem in Distributed Systems does it solve?

Q6 How does Consul handle reliability and high availability?

Q2 What are the core components or concepts in Consul?

Q7 What observability does Consul provide out of the box?

Q3 How does Consul compare to its main alternatives?

Q8 What are common performance bottlenecks in Consul?

Q4 How do you get started with Consul for a new project?

Q9 What security best practices apply when running Consul?

Q5 What configuration options most affect Consul's behavior in production?

Q10 What mistakes do teams commonly make when adopting Consul?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Consul is a tool or platform that

Mitigation

Consul is a tool or platform that automates or simplifies a specific concern in the Distributed Systems domain, reducing manual effort and operational complexity for engineering teams.

A6 Consul provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

Availability

A2 Consul has key building blocks that work

Primitives

Consul has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Consul exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

Information

A3 Consul trades off simplicity against feature richness

Hardening

Consul trades off simplicity against feature richness differently than its alternatives. Choose Consul when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

Performance

A4 Install Consul via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

Gotchas

A9 Run Consul with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

Assessment

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

Concurrency

A10 Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.

Documentation